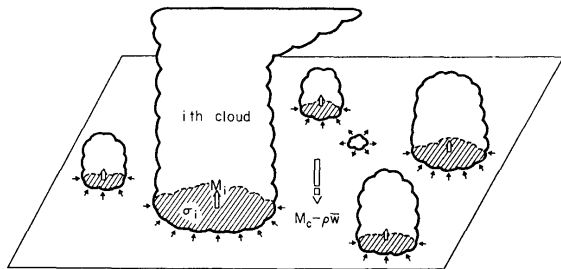


Interaction of a Cumulus Cloud Ensemble with the Large-Scale Environment, Part I



Collective effect of clouds

Rising mass flux, compensating subsidence and detrainment change environmental heat and moisture.

$$M_c = \sum_i \rho \sigma_i w_i$$

σ_i : area fraction.

w_i : vertical velocity.

The theory couples a cumulus population to its large-scale environment

A spectrum of entraining plumes

Each cloud type has a fractional entrainment rate λ .

More entrainment dilutes the plume.
In this sounding, its top is lower.

The remaining unknown is $\mathcal{M}_B(\lambda)$:
the mass flux allocated to each type at the mixed-layer
top.

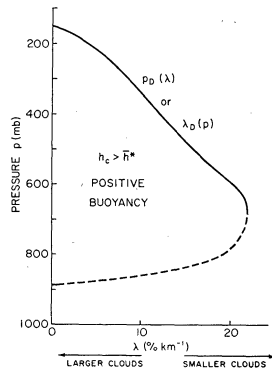
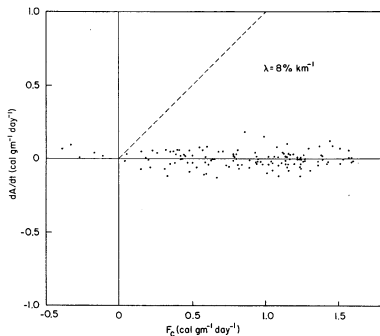


Fig. 8: cloud-top pressure vs. entrainment.

The environment fixes the vertical structure of each cloud type



Marshall Islands: observed dA/dt is small relative to the forcing.

Cloud work function $A(\lambda)$

An integral measure of plume buoyancy. Clouds adjust faster than the large-scale forcing changes.

$$\int K(\lambda, \lambda') \mathcal{M}_B(\lambda') d\lambda' + F(\lambda) \simeq 0$$

Cloud stabilization balances the forcing. This equation constrains $\mathcal{M}_B(\lambda)$ for active cloud types.

Quasi-equilibrium provides the closure for the ensemble's mass flux

What is missing for a 2D thermal field

4

No distributions reported for

Thermal size at a specified height.

Thermal positions in the (x, y) plane.

Distances between thermals.

The closure determines mass flux. Cloud number and lifetime remain separate unknowns.

Possible use: our inference

If the flux closure is solved, Eq. (145) gives a conditional route to cloud counts:

$$\mathcal{N}(\lambda) = \frac{\mathcal{M}_B(\lambda) \tau(\lambda)}{m_B(\lambda)}$$

Needs lifetime τ , mass transported per cloud m_B , and then a cloud-thermal mapping.

Even a known cloud count would leave thermal positions and spacing open